



3.4 The unit circle and radian measure

Geometry and trigonometry · IB Diploma · Concept guide and original exam-style practice

Understand the concept

An angle of 1 radian is subtended at the centre of a circle by an arc equal in length to the radius; π radians = 180° . On the unit circle (radius 1, centred at origin), for angle θ measured anticlockwise from the positive x-axis, $\cos\theta$ and $\sin\theta$ are the x- and y-coordinates of the corresponding point. Arc length $s=r\theta$ and sector area $A=(1/2)r^2\theta$, with θ in radians.

Worked example: follow the reasoning

A circle has radius 8 cm. Find the arc length and sector area for a sector with angle $5\pi/6$ radians.
Arc length $s = r\theta = 8(5\pi/6) = 40\pi/6 = 20\pi/3 \approx 20.9$ cm.
Sector area = $(1/2)r^2\theta = (1/2)(64)(5\pi/6) = 32(5\pi/6) = 160\pi/6 = 80\pi/3 \approx 83.8$ cm².

Quick check · multiple choice

What is 135° converted to radians?

- A $\pi/4$
- B $3\pi/4$
- C $2\pi/3$
- D $5\pi/6$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A sector of a circle has radius 6 cm and arc length 10 cm. [7]
- (a) Find the angle of the sector in radians.
 - (b) Find the area of the sector.
 - (c) On the unit circle, state the exact coordinates of the point corresponding to angle $7\pi/6$, and hence state the value of $\cos(7\pi/6)$.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $s=r\theta \rightarrow 10=6\theta \rightarrow \theta=10/6=5/3$ radians.

(b) $\text{Area}=(1/2)r^2\theta=(1/2)(36)(5/3)=(18)(5/3)=30 \text{ cm}^2$.

(c) $7\pi/6 = \pi + \pi/6$, in the third quadrant. Reference angle $\pi/6$: $\cos(\pi/6)=\sqrt{3}/2$, $\sin(\pi/6)=1/2$. In the third quadrant both cos and sin are negative: point $(-\sqrt{3}/2, -1/2)$. So $\cos(7\pi/6) = -\sqrt{3}/2$.

Quick check answer

B. $135^\circ \times \pi/180 = 135\pi/180 = 3\pi/4$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.